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RESEARCH PROFILE

I am a biological anthropologist and bioarchaeologist investigating how demographic, environmental, and social change shaped human health, behaviour, and biological variation in past populations. My previous work has examined body size and proportions across ancient China, with a particular focus on how ecological conditions, including climate and altitude, interacted with cultural and demographic processes, such as subsistence strategies and population mobility, to shape human development and adaptations over long timescales. Extending this broader interest in how large-scale ecological and social processes become embodied, I am now particularly interested in how population pressure, environmental instability, and increasing social complexity influenced interpersonal violence and other embodied outcomes. Methodologically, I combine osteological analysis with quantitative and computational approaches, including Bayesian modelling, spatiotemporal analysis, and machine learning, to test competing explanations for large-scale patterns in the archaeological record.

RESEARCH INTERESTS

- Biological anthropology, bioarchaeology, and human skeletal biology
- Human health, growth, development, and adaptation under long-term environmental and social change
- Population dynamics, environmental instability, social inequality, and interpersonal violence in past societies
- Quantitative and computational bioarchaeology, including Bayesian inference and spatiotemporal modelling
- High-altitude adaptation, growth, and health in past populations
- Commingled human remains, secondary burial practices, and mortuary formation processes

ACADEMIC APPOINTMENTS

2026-present	Research Assistant Professor, Institute for the Humanities and Social Sciences, The University of Hong Kong Research in biological anthropology, osteoarchaeology, quantitative/computational archaeology, high-altitude adaptation, and demography and health in past populations.
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EDUCATION

2021-2025 Cambridge, UK	PhD in Archaeology / Biological Anthropology, University of Cambridge Department of Archaeology. Dissertation research on high-altitude adaptation, human body size and proportions, and comparative skeletal variation in ancient Tibetan and lowland Chinese populations.
2018-2021 Chengdu, China	MA in Archaeology, Sichuan University School of Archaeology and Museology. GPA: 3.8/4.0. Thesis: Research on human skeletal remains from the Phiyang-Dungkar tombs in Western Tibet.
2014-2018 Chengdu, China	BA in Archaeology, Sichuan University School of Archaeology and Museology. GPA: 3.6/4.0.

RESEARCH GRANTS AND FUNDING

2026	URC Seed Fund for Basic Research for New Staff, The University of Hong Kong Principal Investigator; HK\$150,000. Project: The Cost of Complexity: Modelling Inequality in Growth and Adult Mortality across Long-Term Social Transitions in China's Central Plains
2021-2025	Harding Distinguished Postgraduate Scholars Programme, University of Cambridge Full PhD scholarship and research support.
2022	International Committee of the Red Cross (ICRC) Research Scholarship Research scholarship supporting work related to missing persons and unidentified human remains.
2018-2021	National Scholarship for Master's Degree, China First Class.

2020	Outstanding Student Leader Award, Graduate School, Sichuan University
2018	Outstanding Undergraduate Dissertation, Sichuan University
2018	China Guardian & Xu Bangda Art Scholarship
2017	Outstanding Undergraduate Student Award, Sichuan University

PUBLICATIONS

Journal articles

- Cao, D.**, Crema, E. R., & Pomeroy, E. (2026). Multifactorial drivers of spatial variation in human body form across modern China. *Scientific Reports*. <https://doi.org/10.1038/s41598-026-57537-8>
- Wang, X., Yang, A., Yang, F., Yuan, H., **Cao, D.**, Li, J., Huo, W., Lu, S., & Li, S. (2026). Subsistence change in Iron Age to Tubo-period western Xizang: A stable isotope study from Phiyang Dunkar. *Journal of Archaeological Science: Reports*, 70, 105639. <https://doi.org/10.1016/j.jasrep.2026.105639>
- Cao, D.**, Liu, X., Gu, W., Zhu, H., Zhang, R., Zhou, Z., Wei, Q., Zou, J., Qiu, Y., Chen, J., Ding, L., Pomeroy, E., & Yuan, H. (2025). Dental caries as indicators of agricultural practices in the foothills of Neolithic China. *Archaeological Research in Asia*, 43, 100645. <https://doi.org/10.1016/j.ara.2025.100645>
- Wang, X., **Cao, D.**, Gu, W., Wei, Q., Chen, X.-L., Zhang, R., Ding, L., Tang, Z., Teng, X., Wang, T., Zou, J., Qiu, Y., Fu, Q., & Yuan, H. (2025). Multi-isotope analysis reveals human millet-based diets and limited mobility in the Central Plains of China ca. 5000 years ago. *International Journal of Osteoarchaeology*, 35, e3407. <https://doi.org/10.1002/oa.3407>
- Cao, D.**, Crema, E. R., & Pomeroy, E. (2024). Estimating intralimb proportions for commingled remains. *International Journal of Osteoarchaeology*. <https://doi.org/10.1002/oa.3326>
- Yi, B., Yuan, H., Wang, T., **Cao, D.**, Yang, Y., & Hu, Y. (2024). Life history of a high-status noblewoman from the late Shu state during the Eastern Zhou period (770-221 BC): Childhood stress and adult stability. *International Journal of Osteoarchaeology*, 34(1).
- Tao, L., Yuan, H., Zhu, K., Liu, X., Guo, J., Min, R., He, H., **Cao, D.**, Yang, X., Zhou, Z., Wang, R., Zhao, D., Ma, H., Chen, J., Zhao, J., Li, Y., He, Y., Suo, D., Zhang, R., Li, S., Li, L., Yang, F., Li, H., Zhang, L., Jin, L., & Wang, C.-C. (2023). Ancient genomes reveal millet farming-related demic diffusion from the Yellow River into southwest China. *Current Biology*, 33(22), 4995-5002.e7. <https://doi.org/10.1016/j.cub.2023.09.055>
- Lu, H., Li, Z., Ciren, C., **Cao, D.**, Gao, X., & Li, L. (2022). Sding Chung: An early multi-burial cave site on the Tibetan Plateau. *Antiquity*, 96(387), 745-753. <https://doi.org/10.15184/aqy.2022.42>
- Yuan, H., Gu, W., Wei, Q., Wu, Q., Ding, L., & **Cao, D.** (2022). Analysis of dental caries in the Yangshao population at the Qingtai site, Zhengzhou. *Acta Anthropologica Sinica*, 41(2), 226-237. [in Chinese] <https://www.anthropol.ac.cn/EN/10.16359/j.1000-3193/AAS.2021.0023>
- Cao, D.**, & Yuan, H. (2021). Preliminary analysis of commingled human remains. *Yellow River, Yellow Earth, Yellow Race*, (06), 41-44. [in Chinese]

Newsletter

- Cao, D.** (2017, November 17). Seeing humans through bones: A multidimensional perspective on ancient human life - Summary of the Second Annual Meeting of the Osteoarchaeology Committee of the Chinese Archaeological Society. *China Cultural Relics News*, (006). [in Chinese]

Thesis

- Cao, D.** (2025). Adaptation at High Altitudes: A Comparative Analysis of Body Size and Proportions in Ancient Tibetan and Lowland Chinese Populations [Apollo - University of Cambridge Repository]. <https://doi.org/10.17863/CAM.123688>
- Cao, D.** (2021). Research on human skeletal remains from the Phiyang-Dungkar tombs in Western Tibet. Master's thesis, Sichuan University, Chengdu, China. [in Chinese] <https://kns.cnki.net/KCMS/detail/detail.aspx?dbname=CMFDTEMP&filename=1022445375.nh>

CONFERENCE PRESENTATIONS

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| 2026 | Cao, D. , Yuan, H., & Pomeroy, E. <i>Growth and health in high-altitude environments: Body form and pathology in ancient Tibetan Plateau populations</i> . International Symposium on Human Osteoarchaeology: Disease, Health and Ancient Society, School of Archaeology and Cultural Heritage, Zhengzhou University, Zhengzhou, China, 7 July 2026. (Podium) |
| 2026 | Cao, D. , Lü, H., Yang, F., Li, S., Yuan, H., Ciren, C., Wangdue, S., He, Y., Li, Y., Liu, X., Ji, K., Zou, J., & Pomeroy, E. Adaptation or constraint? Skeletal evidence for growth and health in Himalayan highland populations. Trans-Eurasian Highlands: Insights from Multidisciplinary Archaeological Research, Center for Archaeological Science, Sichuan University, Chengdu, China, 9-13 March 2026. (Podium) |
| 2026 | Cao, D. , Pomeroy, E., Lü, H., Yuan, H., Yang, F., & Li, S. Bayesian modelling reveals spatiotemporal patterns of body size variation in Holocene Chinese populations. The |

Second Early China Development and Its Long-Term Impact Workshop, The University of Hong Kong, Hong Kong. (Podium)

- 2025 **Cao, D.**, Crema, E. R., Lü, H., Yuan, H., & Pomeroy, E. Tracking spatial and temporal variation in body form across ancient Chinese populations. Annual Conference of Computer Applications and Quantitative Methods in Archaeology (CAA UK), Cambridge, United Kingdom. (Podium)
- 2025 **Cao, D.**, Crema, E. R., Lü, H., Yuan, H., & Pomeroy, E. Bayesian models reveal spatiotemporal patterns of body size variation in Holocene Chinese populations. Annual Meeting of the Human Osteoarchaeology Committee, Chinese Archaeological Society, Yili, Xinjiang, China. (Podium)
- 2025 **Cao, D.**, Crema, E. R., & Pomeroy, E. Tracking long-term and geographic variation in body size among Chinese populations: Combined insights from contemporary and archaeological evidence. 94th Annual Meeting of the American Association of Biological Anthropologists, Baltimore, USA. (Podium)
- 2024 **Cao, D.**, Pomeroy, E., Lu, H., Yuan, H., Yang, F., & Li, S. An analysis of commingled human remains from two Bronze to Iron Age sites in West Tibet. 30th Annual Meeting of the European Association of Archaeologists, Rome, Italy. (Podium)
- 2024 **Cao, D.**, Liu, X., & Yuan, H. Dental caries as indicators of agricultural practices in the foothills of Neolithic China. 24th European Meeting of the Paleopathology Association, Leiden, Netherlands. (Poster)
- 2023 **Cao, D.**, Crema, E. R., & Pomeroy, E. Estimating intra-limb proportions for commingled remains. 23rd Annual Conference of the British Association for Biological Anthropology and Osteoarchaeology, London, UK. (Poster)
- 2023 **Cao, D.**, Lu, H., Yang, F., Li, S., Wangdui, X., Yuan, H., & Pomeroy, E. Bioarchaeological insights into health and cultural practices on the ancient Tibetan Plateau in Central Asia. 29th Annual Meeting of the European Association of Archaeologists, Belfast, UK. (Poster)
- 2019 **Cao, D.**, & Yuan, H. Life-course study of Zhao Xin, an official of the Northern Qi Dynasty. Third Annual Meeting of the Human Osteoarchaeology Committee, Chinese Archaeological Society, Qingdao, China. (Poster)
- 2018 **Cao, D.** Dental caries in the Late Neolithic Gaoshan population in southwest China. Osteoarchaeology Session, Second Chinese Archaeological Congress, Chengdu, China. (Podium)

FIELDWORK EXPERIENCE

Excavations

- 2021 **Sdingchung Cave Site, Zhongba, Xizang (Tibet), China**
Participated in excavation and was responsible for analysis of human skeletal remains.
- 2016 **Haimenkou Site, Jianchuan, Dali, Yunnan, China**
Participated in excavation and artefact sorting.

Surveys and skeletal documentation

- 2026 **Jiaoliuqiu Cave, Guiyang, Guizhou, China**
Field survey.
- 2025 **Shuanghe Cave, Suiyang, Zunyi, Guizhou, China**
Field survey.
- 2020 **Shidaqiu Site, Maerkang, Aba, Sichuan, China**
Field survey and in situ sorting of human skeletal remains.
- 2019 **Xincun Site, Wuding, Chuxiong, Yunnan, China**
Documentation and sorting of human skeletal remains.
- 2019 **Ladian Site, Yuanmou, Chuxiong, Yunnan, China**
Field survey and in situ sorting of human skeletal remains.
- 2019 **Jiangxifen Site, Wuding, Chuxiong, Yunnan, China**
Documentation and sorting of human skeletal remains.
- 2018 **Zhijiaozhongxin Site, Fenyang, Shanxi, China**
Field survey, documentation, and collection of human skeletal remains in situ.

2018

Qingtai Site, Zhengzhou, Henan, China

Field survey, documentation, and sorting of human skeletal remains.

TEACHING AND SUPERVISION

2024-2025	Supervisor, From Data to Interpretation, Department of Archaeology, University of Cambridge Coordinator: Dr Enrico R. Crema. Supervised students in data analysis tools, including R, with emphasis on archaeological and anthropological datasets.
2023-2024	Supervisor, Comparative Human Biology, Department of Archaeology, University of Cambridge Coordinator: Dr Mark Dyble. Supervised topics in comparative human biology, evolutionary adaptation, life history, and socio-cultural factors influencing human variability and health.
2023-2025	Demonstrator, Decoding the Skeleton, Department of Archaeology, University of Cambridge Coordinator: Dr Emma Pomeroy. Demonstrated skeletal analysis techniques, including osteological sorting, age and sex estimation, anthropometrics, and palaeopathological observation.
2024-2025	Demonstrator, Archaeology in Action, Department of Archaeology, University of Cambridge Coordinator: Dr Simon Stoddart. Demonstrated techniques in sorting and analysing human skeletal remains.
2018-2021	Student Teaching Assistant, Sichuan University Bioarchaeology Laboratory, Center for Archaeological Science Assisted with laboratory management, equipment maintenance, sample sorting, and teaching.

PUBLIC ENGAGEMENT AND ADDITIONAL EXPERIENCE

2021	Speaker, TEDxKids@Chengdu “What I Found from Buried Ancient Teeth / 一颗蛀牙来认识古蜀先民”.
2019	Sichuan University-Harvard University Study Camp on Ancient Shu Civilisation Funded by the Fairbank Center for Chinese Studies at Harvard University and the Research Center for Western China at Sichuan University.

SKILLS AND SOFTWARE

- Osteological analysis: skeletal sorting, demographic reconstruction, commingled remains, age/sex estimation, anthropometrics, and palaeopathological observation.
- Statistical and computational analysis: R, Bayesian modelling, data visualisation, and reproducible research workflows.
- Spatial analysis: qGIS, basic map making.
- 3D imaging and morphometrics: Artec Studio, 3D Slicer, SlicerMorph, MeshLab, ImageJ.
- Languages: Chinese (native); English (fluent).